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TITLE: AI-DRIVEN DIGITAL HUMAN

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Certificate of Approval

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Abstract

AI-Driven Digital Human is designed to revolutionize the news industry system through Artificial Intelligence and cutting-edge technologies. By integrating advanced framework of Unreal Engine our system introduces two modes of chatbot, the Repeater mode and the Interview mode. The Repeater mode is responsible for delivering the news content based on the input given to the system by the user whereas the Conversational mode acts as a real news host conducting real-time interviews. The key element of this project is 3D human-like Avatar with features like blinking, lifelike movement and natural speech synthesis using MetahumanSDK and ConvAI.

In conclusion, AI-Digital Human Newscaster revolutionize the traditional news industry system into modern system by eliminating gender biasness, introducing customization of avatar selection and enhanced user experience. The aim of this project is to focus on increasing user engagement by conducting real-time interviews irrespective of the gender selection.

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Chapter 1

This chapter includes a comprehensive introduction about our project AI-Driven Digital Human. As AI is revolutionizing every industry our project aims to build a bridge between the news industry and users by introducing 3D-Avatars.

Introduction

In an era where news is delivered through digital channels The AI-Driven Digital Human project presents an initiative to revolutionize the news delivery to the audience by harnessing the power of cutting-edge technologies. The system is designed to offer two distinctive chatbot modes ensuring the user engagement as well as conducting real-time interviews all while eliminating biasness by allowing the user to select the gender of the avatar. The core of the project lies in the 3D Digital Model to present the news with the mission to engage users, reduce gender biasness and offer a transformative experience by introducing different dialects to build the gap between humans and AI Machines.

1.1. Problem Statement

The AI based digital news caster is a revolutionary system that utilizes advanced AI and incorporates two modes of chatbot to enhance user engagement as well as take real-time interviews while eliminating biasness. The objective is to produce a 3D avatar that uses speech synthesis technology to produce natural-sounding speech just like a news anchor. Our AI based project aims to address the limitations of news delivery through development of an AI-based digital newscaster with different dialects and modes of chatbot.

1.2. Purpose

The primary purpose of the AI-Driven Digital Human project is to transform traditional paradigms of news delivery and offer following key objectives:

- **The Repeater Mode:**

This mode is used to engage users in a more dynamic way, acting as a news anchor and reading the input as a news script while delivering news content in an efficient way. The input can be given in textual form or recorded audio.

- **The Interview Mode:**

This mode engages users in a more dynamic way, acting as an interviewer and asking back-to-back questions. This mode acts as a real news host interviewing the user and having a human like conversation.

- **Personalization of Gender of the Newscaster:**

Users are allowed to choose the gender of the avatar for personalization experience and for delivering unbiased news and interviewing in a fair and impartial manner regardless of the gender. Our model includes 4 3D avatars, Stephane, Malika, Payton, and Gavin to ensure that regardless of the selected gender all avatars maintain an impartial tone, guaranteeing fairness in news delivery.

- **Levels of Avatar:**

There are 2 levels of avatar selection which differ in diverse range of features and basic user widgets. Users are allowed to choose the respective level which can be activated using different triggers.

- **Enhanced User Engagement:**

The 3D Human Model with its distinctive chatbot modes the “Repeater Mode” and the “Interviewer Mode” ensures user interaction and engaging news discourse.

- **Multiple Dialects:**

The 3D Human Model includes seven different dialects for user interaction to ensure inclusivity and to cater to a global user base.

- **Multifaceted Operational Modes:**

The AI-Driven Digital Human includes two operational modes Parallel and Sequential Mode. The parallel mode allows the user to engage with multiple tasks simultaneously whereas the sequential mode offers a linear

progression of tasks.

- **Deliver Unbiased News:**

The user is allowed to choose the desired gender of the news anchor and regardless of the chosen gender the digital models will maintain a neutral and unbiased tone, eliminating any gender-related stereotypes which will ensure fair news delivery.

- **Versatile Utility:**

Beyond the role of The Digital Human Model as a newscaster it can act as various entities including a Medical Advisor, Educational Instructor, or a Language Tutor upon integrating with the appropriate chatbot model.

- **Industry, Innovation, and Infrastructure:**

The project contributes to the news industry by revolutionizing it and setting a precedent for other media organizations to adapt and enhance their delivery of the contents. The project aligns with the Sustainable Development Goals of gender equality by allowing the user to select the desired gender of the news anchor and through innovative news delivery it modernizes the news industry and contributes to decent work and economic growth.

1.3. Product Scope

The scope of the project lies in the development of the 3D Digital Human Model which offers an immersive experience of unbiased consumption of news regardless of the gender chosen by the user. The project further extends to two primary modes of the chatbot, the “Interview” mode which facilitates real-time user interaction and the "Repeater” mode which replicates the user input. Ultimately, the project scope of the AI-Driven Digital Human is to establish an innovative news broadcasting digital model to not only deliver unbiased news but also cater to the dynamic needs and expectations of the Pakistani audience.

Chapter 2

In this chapter we will be defining our project related past contributions considering it specific to our project, needed for research purposes including past models and published/unpublished research papers.

Literature Survey

The proliferation of artificial intelligence (AI) technologies has engendered a paradigm shift across numerous sectors, precipitating a surge of scholarly inquiry into its manifold implications. Among these, the intersection of AI and journalism stands as a focal point of examination, as evidenced by the seminal work "Artificial Intelligence in the News: How AI Reshapes Journalism and the Public Arena" (2024). Authored by a team of researchers, this publication illuminates the transformative potential of AI within journalistic practices and its consequential impact on the broader societal discourse.

Published on March 6, 2024, this journal article serves as a poignant testament to the burgeoning discourse surrounding AI's role in journalism. Notably, the authors elucidate the impending release of revolutionary AI technologies, which have elicited considerable intrigue and scrutiny from experts spanning diverse fields. Central to their investigation is an exploration of the motivations driving organizations to integrate AI-based systems into their operational frameworks, thereby heralding a paradigmatic shift away from traditional methodologies.

In their pursuit of insights, the authors employ a qualitative approach, conducting interviews to glean firsthand perspectives on the burgeoning reliance on AI within journalistic endeavors. Through this empirical lens, they discern emergent trends and evolving dynamics, shedding light on the intricate interplay between AI technologies and the journalistic landscape. Such endeavors underscore the imperative of scholarly engagement with the transformative potential of AI, not only within journalism but across

multifarious domains.

As the literature surrounding AI and journalism continues to burgeon, the present study endeavors to contribute to this burgeoning discourse by synthesizing existing scholarship and advancing novel insights into the evolving relationship between AI technologies and journalistic practices. Through a comprehensive review of pertinent literature, this paper seeks to elucidate the nuanced contours of this dynamic intersection, thereby enriching our collective understanding of AI's impact on the dissemination of news and its broader ramifications for societal discourse.

In recent years, the pervasive influence of Artificial Intelligence (AI) has undergone remarkable strides, catalyzing numerous advancements in the realm of AI-driven chatbots. These developments have not only revolutionized various industries, including media [8], but have also paved the way for innovative product developments. Noteworthy among these advancements is the creation of the world's first female AI news anchor, a pioneering achievement attributed to Chinese developers, notably Xin Xiaomeng [1]. This seminal creation, as chronicled in an article featured in "The Indian Express" in February 2018, epitomizes a watershed moment in the evolution of AI technologies and their integration into journalism.

The emergence of Xin Xiaomeng as a pioneering AI news anchor underscores the profound impact of AI on the broadcasting and media sectors. This groundbreaking innovation symbolizes a paradigmatic shift in the landscape of AI and journalism, signaling the advent of novel methodologies and technological applications that challenge traditional conventions within the news industry.

The significance of Xin Xiaomeng lies not merely in its technological novelty but also in its tangible contributions to the field of journalism. By seamlessly blending AI capabilities with the art of news presentation, Xin Xiaomeng exemplifies the transformative potential of AI-driven solutions in enhancing the efficiency and efficacy of media dissemination.

Moreover, the advent of Xin Xiaomeng raises pertinent questions regarding the viability of traditional news industry systems in the face of rapid technological advancements. As AI-powered innovations continue to redefine the contours of journalism, stakeholders within the traditional news ecosystem are compelled to confront the imperative of adaptation and evolution to remain relevant in an increasingly digitized landscape.

Considering these developments, the seminal creation of Xin Xiaomeng stands as a testament to the profound synergy between AI technologies and journalistic practices. As scholarly discourse surrounding AI and journalism burgeons, the case of Xin Xiaomeng offers a compelling narrative that underscores the transformative potential of AI in redefining the dynamics of media dissemination and shaping the future trajectory of journalism.

In the second article, "ERICA: The ERATO Intelligent Conversational Android," authored by Dylan F. Glas and Takashi Minato, the focus shifts to the pioneering development of an AI-based conversational android system known as ERICA. This system represents a significant leap forward in the realm of interactive AI technology, showcasing the capacity for nuanced conversational engagement with users. Central to ERICA's design is the integration of advanced AI technologies [7], leveraging natural speech synthesis capabilities to emulate lifelike human interaction. The authors' exploration of ERICA underscores the burgeoning interest and investment in AI-driven conversational agents, [2] reflecting a broader trend towards the development of increasingly sophisticated human-like interfaces. By delving into the intricacies of ERICA's design and functionality, Glas and Minato contribute valuable insights to the evolving discourse surrounding AI-human interaction, shedding light on the transformative potential of AI technologies in shaping future modes of communication and interaction.

In the landscape of media innovation, the article titled "AI News Presenters: South Korea & India Lead the Way in Media Innovation," penned by Livemint and published in July 2023, stands as a beacon illuminating the transformative trend of AI-driven news anchors. This seminal work not only delves into the pivotal roles assumed by South Korea and India in spearheading this groundbreaking innovation but also introduces Lisa, an AI-based news anchor developed by a prominent South Korean television network. Lisa's unparalleled capacity for natural speech synthesis heralds a paradigm shift in news production, transcending conventional boundaries and symbolizing a seismic transformation in the traditional media landscape. Through its incisive analysis, Livemint not only underscores the rising prominence of AI technologies in reshaping contemporary journalism but also underscores the imperative for adaptability within an ever-evolving media ecosystem. By shedding light on Lisa's [3] profound impact within the media industry, this article emerges as a significant contribution to the scholarly discourse surrounding the transformative potential of AI in redefining the future trajectory of news dissemination.

In the realm of AI advancements, the scholarly discourse converges on the fourth article titled "Advancements in AI-Blurring the Lines between Humans & Robots," which sheds light on the groundbreaking work of David Hanson. Central to this discourse is Hanson's pioneering efforts in the development of an AI-based news model named "Sophia." This work represents a significant milestone in the evolution of AI technologies, particularly in the domain of neuro-symbolic controllers, aimed at enhancing motor function and fostering human-like interactions. Noteworthy is Sophia's [4] remarkable ability to synthesize natural speech and mimic human emotions and gestures [6], effectively erasing the traditional boundaries that delineate human-robot interactions. Through Hanson's innovative pursuits, this article elucidates the transformative potential of AI in transcending the conventional distinctions between artificial intelligence and human cognition, thereby forging new pathways in the burgeoning field of human-robot interaction.

In the realm of Natural Language Processing (NLP) advancements, the discourse coalesces around the fifth article titled "Advancing Natural Language Processing: The Case of Jais and Jais-chat," authored by an esteemed team of expert publishers and published in 2023. This scholarly work delves into the development and functionality of Jais-chat, an Arabic-centered language model, emblematic of cutting-edge innovations in the field of AI and NLP. With a specialized focus on catering to Arab communities, Jais-chat emerges as a pivotal tool for facilitating effective communication and transcending language barriers. Through its nuanced exploration of Jais-chat's capabilities, this publication [5] underscores the transformative potential of AI and NLP in fostering cross-cultural understanding and facilitating seamless linguistic interactions.

Chapter 3

This chapter provides information about the project and how it is implemented.

Design

This section explains the needs for our system to communicate effectively and to ensure smooth integration.

3.1. User and System Requirements

In this project the user is provided with a friendly interface to interact easily and deliver news. The targeted audience for our project is the news industry and news anchors.

3.2. External Interface Requirements

This section outlines the external interface requirements for our project.

3.2.1. User Interface

- **Compatible browser and Internet Connectivity:**

The user is required to have a compatible browser to access the web application and stable internet connectivity.

- **Login/Signup:**

The user should have valid login details to further access the webpage. If necessary, the user might need to signup if they don't have an account.

- **Interaction options:**

The user after successfully logging in can select the gender of the newscaster and the desire mode of the chatbot to deliver news content.

3.2.2. Hardware Interface

The user requires a fast processor, sufficient memory (RAM), and a powerful graphics card to enable smooth rendering of the digital newscaster's avatar and environment.

3.2.3. Software Interface

This section explains the software requirements of the project.

- **Modes of Chatbot:**

The project introduces two modes of chatbot. The Interview mode of the chatbot uses the API integration of Metahumans and SST for intelligent interaction during interviews. The TTS functionality is used in the Repeater mode to vocalize the news input.

- **3D Human Model:**

3D Avatar is implemented on Unreal Engine using Metahumans frameworks like MetaHumanSDK and ConvAI to create a highly realistic and human-like newscaster to provide user with an immersive experience.

- **Lip Synching and Blinking of the Avatar:**

The avatar has lip sync aligned with the facial expressions and movements. For a more human-like touch the project introduces the eye blinking feature of the avatar using ConvAI and MetahumanSDK.

- **Integration of Chatbot and Avatar:**

The integration of both the chatbots with the 3D model is done through C# Blueprints for seamless and immersive user experience. This integration defines the logic for mapping the chatbot output to trigger specific actions and responses within the 3D-Avatar.

- **Exception Handling:**

In the event of failure or connectivity issues, the model utilizes scripted audio to let the user know of the situation for seamless communication.

- **Natural Speech Sounding:**

The project has used pretrain natural sounding models from Google MetahumanSDK that works best for our project for TTS.

3.2.4. Communication Interface

The user requires a compatible device to access the web application. Basic

communication requires a strong internet connection. The user can communicate with the application after successfully login with a valid email and password. The main communication will be held between the news industry user and the application to deliver reliable news content.

3.3. Functional Requirements

Below is the list of functional requirements for our project. Every functional requirement is listed on a separate table.

3.4. Functional requirements with traceability information

3.4.1. Functional Requirement 1

FR-01: Below is our first functional requirement. In this user is redirected to his/her account upon successful login to access his/her account. If the account is not created, he/she will first create one for successful authorization process later.

Table 2: Table for Functional Requirement 1

Requirement ID	FR-03	Requirement Type			Functional	Use Case #		UC-01	
Status	New	X	Agreed-to	-	Baselined	-	Rejected	-	
Parent Requirement #	FR-01, FR-02								
Description	User is allowed to select the required chatbot modes. The interview mode engages in conversational interactions whereas the repeater mode delivers real-time input through avatar.								
Rationale	User is able to interact by providing real-time input and engage in meaningful conversations for immersive experience.								
Source					Source Document	-			
Acceptance/Fit Criteria	Users can successfully interact with the 3D-Human Avatar.								
Dependencies	None								
Priority	Essential	X	Conditional	-		Optional	-		
Change History	None								

3.4.2. Functional Requirement 2

FR-02: Below is our second functional requirement. The user is able to successfully select the gender of the avatar by triggered actions based on his/her requirement.

Table 3: Table for Functional Requirement 2

Requirement ID	FR-01	Requirement			Functional	Use Case #			UC-01
Status	New	X	Agreed-	-	Baseline	-	Rejected	-	
Parent Requirement #	None								
Description	Users must sign up/login into their accounts to access the system.								
Rationale	Prevention against unauthorized access.								
Source					Source Document		-		
Acceptance/Fit Criteria	User is successfully redirected to the dashboard.								
Dependencies	None								
Priority	Essential	X	Conditional	-	Optional	-			
Change History	None								

3.4.3. Functional Requirement 3

FR-03: Below is our third functional requirement. The user is able to successfully select the modes of the chatbot. The interview mode is to engage in conversational interactions whereas the repeater mode is to deliver real-time input through conversation.

Table 4: Table for Functional Requirement 3

Requirement ID	FR-02	Requirement Type			Functional	Use Case #		UC-01	
Status	New	X	Agreed-to	-	Baselined	-	Rejected	-	
Parent Requirement #	FR-01								
Description	User can select the gender of digital human avatar.								
Rationale	Prevention against gender biasness.								
Source					Source Document	-			
Acceptance/Fit Criteria	User can successfully interact with the chatbot.								
Dependencies	None								
Priority	Essential	X	Conditional	-	Optional	-			
Change History	None								

3.5. Non-Functional Requirements

Non-Functional Requirements for our project are as under:

- **Usability:**

The system has a user-friendly interface to ensure easier usability and engagement of the user.

- **Security:**

To prevent unauthorized access to an application the system includes a login page and a signup page which requires valid email to access the application.

- **Reliability:**

The system is consistent with the required functions and perform them without failure.

- **Scalability:**

The system is able to manage a scalable number of users while ensuring high level performance.

- **Compatibility:**

The project requires compatible desktops for seamless interaction and delivery of the news content.

3.6. Performance Requirements

AI-Driven Digital Newscaster is a web-based application that is deployed on Unreal Engine to provide users with a seamless interface and easier interaction. The project allows real-time interaction with quick responsive conversations with the chatbot. The performance benchmark is dependent on the user's hardware component's compatibility.

Chapter 4

The following chapter outlines the methodology and implementation of our system to address the approach and proposed solutions.

Proposed Solution

This section provides an in-depth exploration of the methodology and the implementation.

4.1. Methodology

To ensure effectiveness of our system's chatbot, different models were used to evaluate the effectiveness of available intelligent chatbot models. Our research was focused on identifying the chatbot model that could provide relevant and appropriate answers to maintain efficient conversations.

After evaluating several chatbot models, to assess their limitations and capabilities, below are the results.

- **DialoGPT:** This model lacked in depth understanding.
- **GPT-3 Turbo:** This model generates biased and irrelevant content without proper insight.
- **Med Llama:** Med Llama provides plausible answers but lacks factual accuracy.
- **Metahumans:** Metahumans generates efficient responses but lacks personalization tailored to the news industry.

4.2. Implementation

The AI-Driven Digital Human Newscaster aims to revolutionize the news industry by using 3D-Avatar with human-like features and natural speech to provide users with an immersive experience. We have utilized Unreal Engine's Metahuman frameworks like MetaHumanSDK, ConvAI and blueprints to handle exceptions. To ensure immersive experience, real human-like features and blinking of eyes we have employed ConvAI and to efficiently enable lip syncing and natural motions MetahumanSDK has been implemented. Additionally, integration of TTS and STT is also implemented to enable the two modes of the chatbot, the repeater and the

interviewer mode to bridge the gap between the user and the Avatar. The interview mode enables the user to engage in immersive interviews with the news avatar whereas the repeater mode provides real-time news delivery based on the user input. Our system ensures seamless integration and user interaction by allowing different dialects option and multifaceted operational mod. The utilizing of UserWidget framework of Unreal Engine for graphical interface is triggered by user's actions and provides smooth communication.

Several companies have ventured into the development of AI-driven avatars, but our system differs due to several reasons and advancements as follow:

- **Real-Time Interviews**

Existing systems focus only on pre-programed news content delivery whereas our model takes real time input and delivers news content. Our system introduces an interview mode which conducts real-time interviews just like a real news anchor.

- **Interactive Chatbot Modes**

Existing systems lack user interaction functionalities, and our system incorporates different modes of chatbots for active user participation as well as allowing the user to select different dialects and operational mode.

- **Gender Customization and levels of avatar**

Our model allows the user to select the desired level of the avatar with different features in each level for customization experience. The user is also allowed to select the desired gender and avatar to ensure unbiased news delivery.

Chapter 5

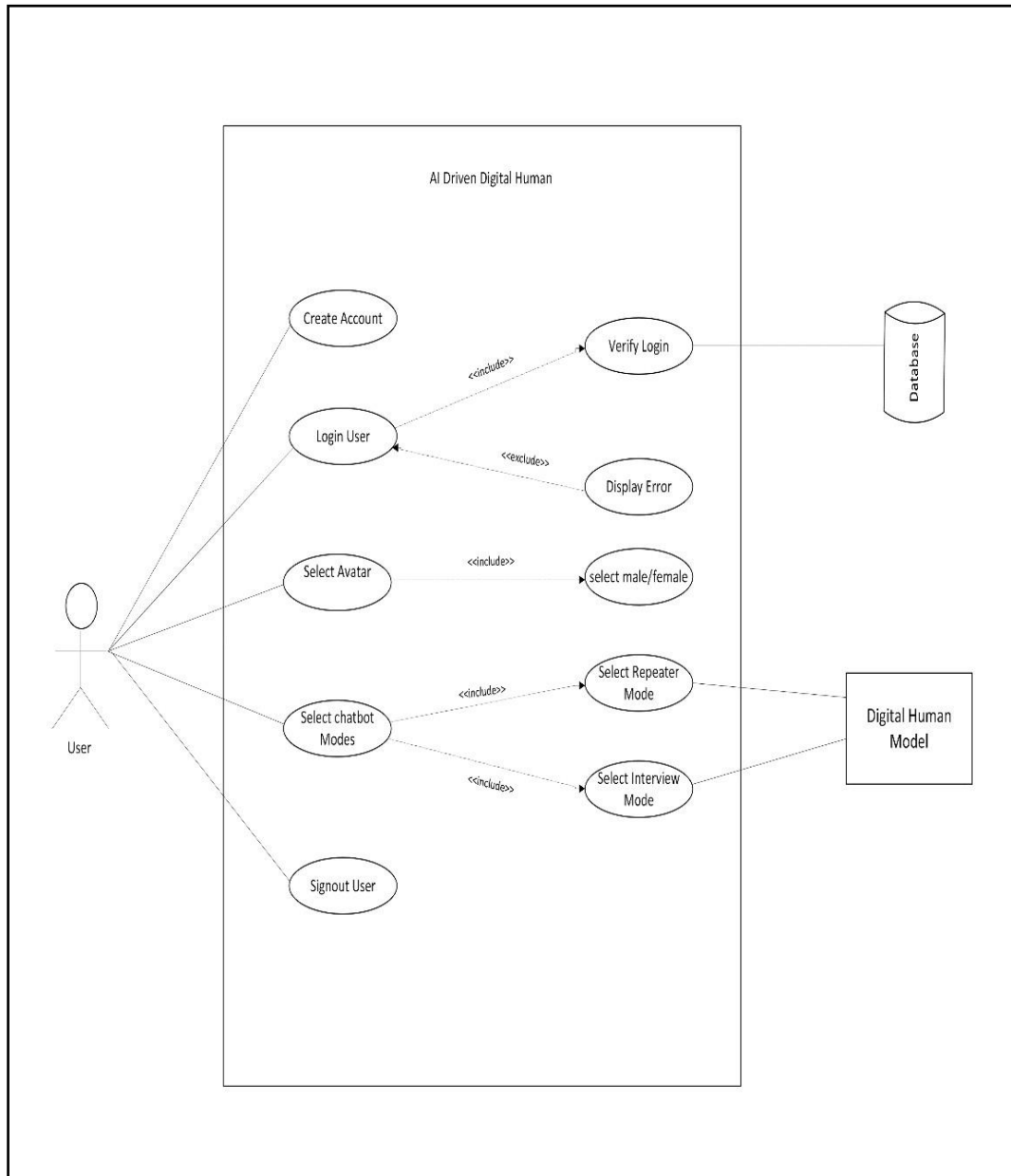
This chapter includes the project's design and architecture.

Project Design/Architecture

5.1. Use Case View

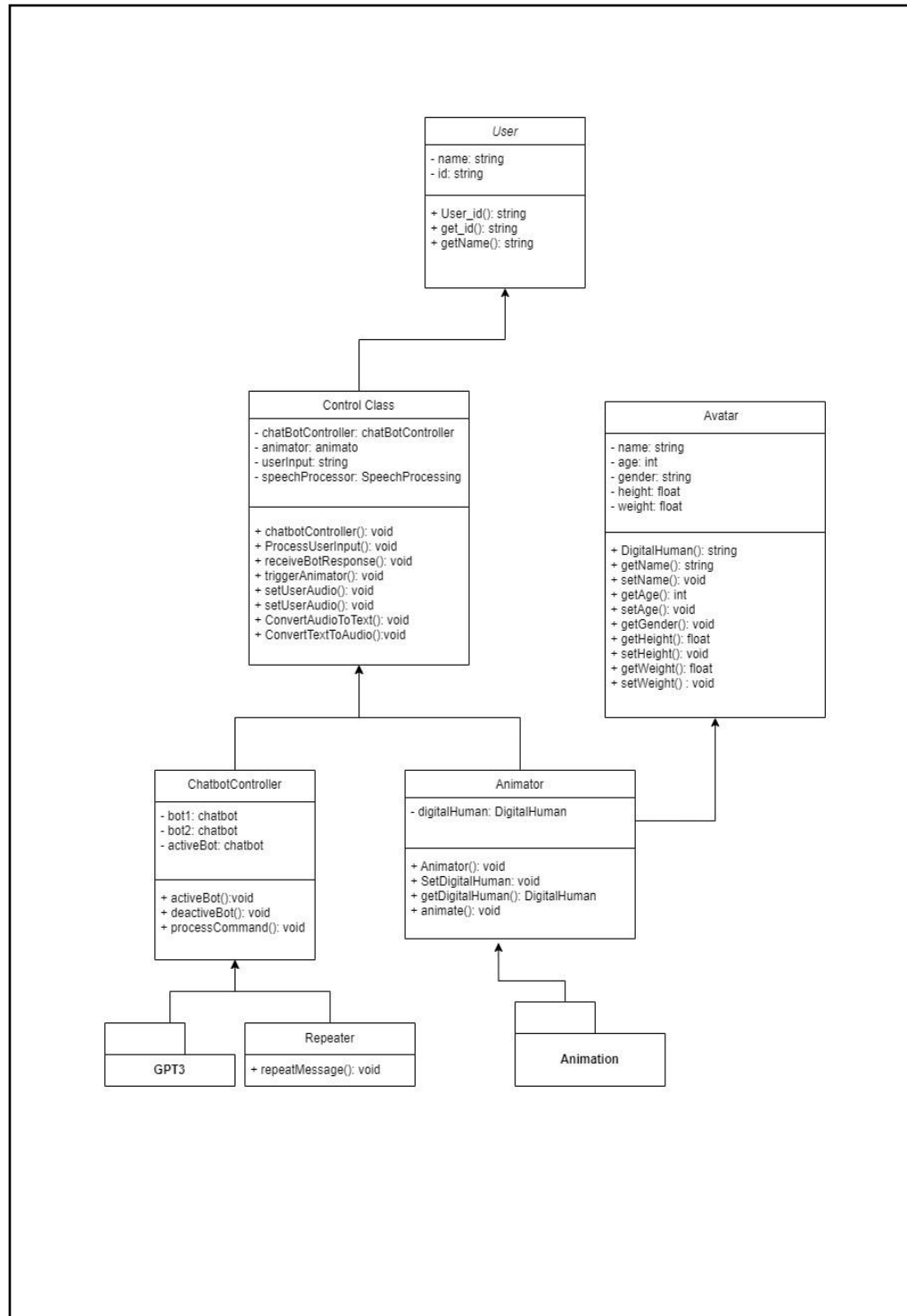
5.1.1. UC-01

Use Case View



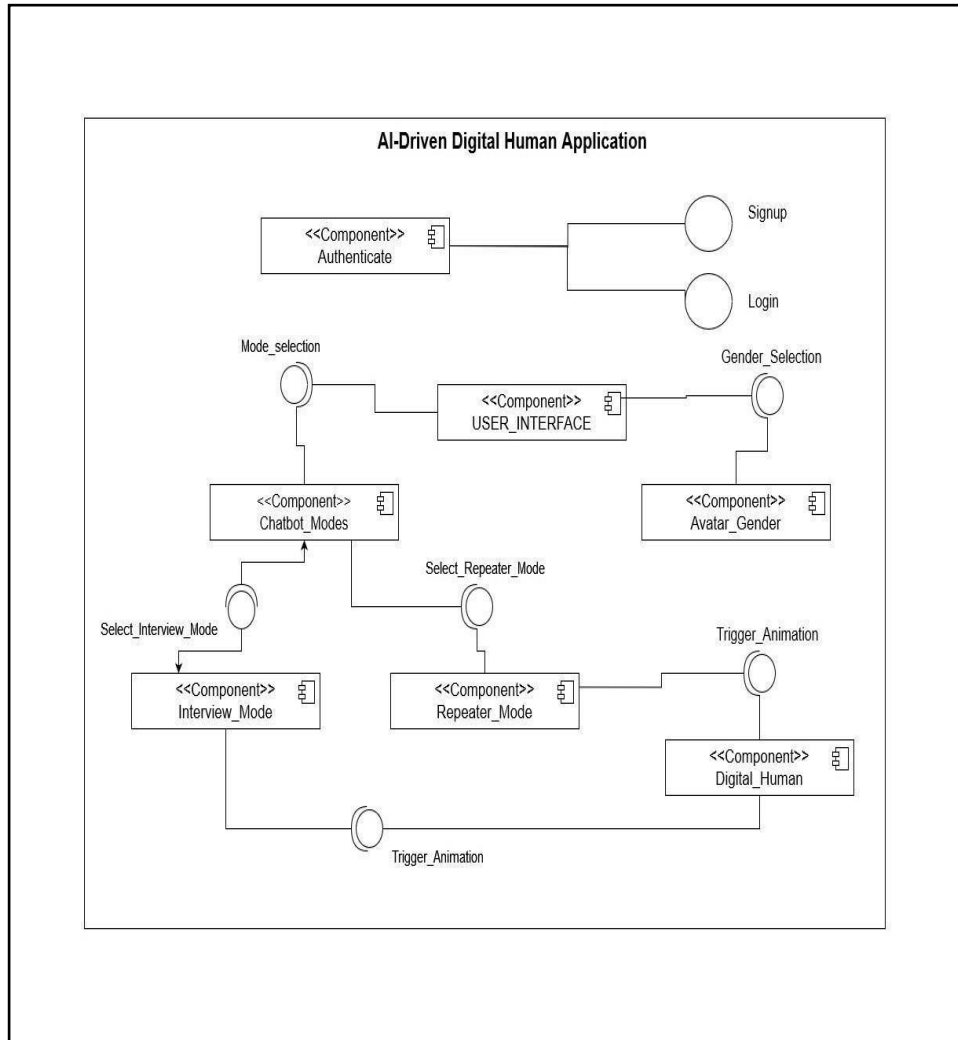
5.2. Logical View

5.2.1. Class Diagram



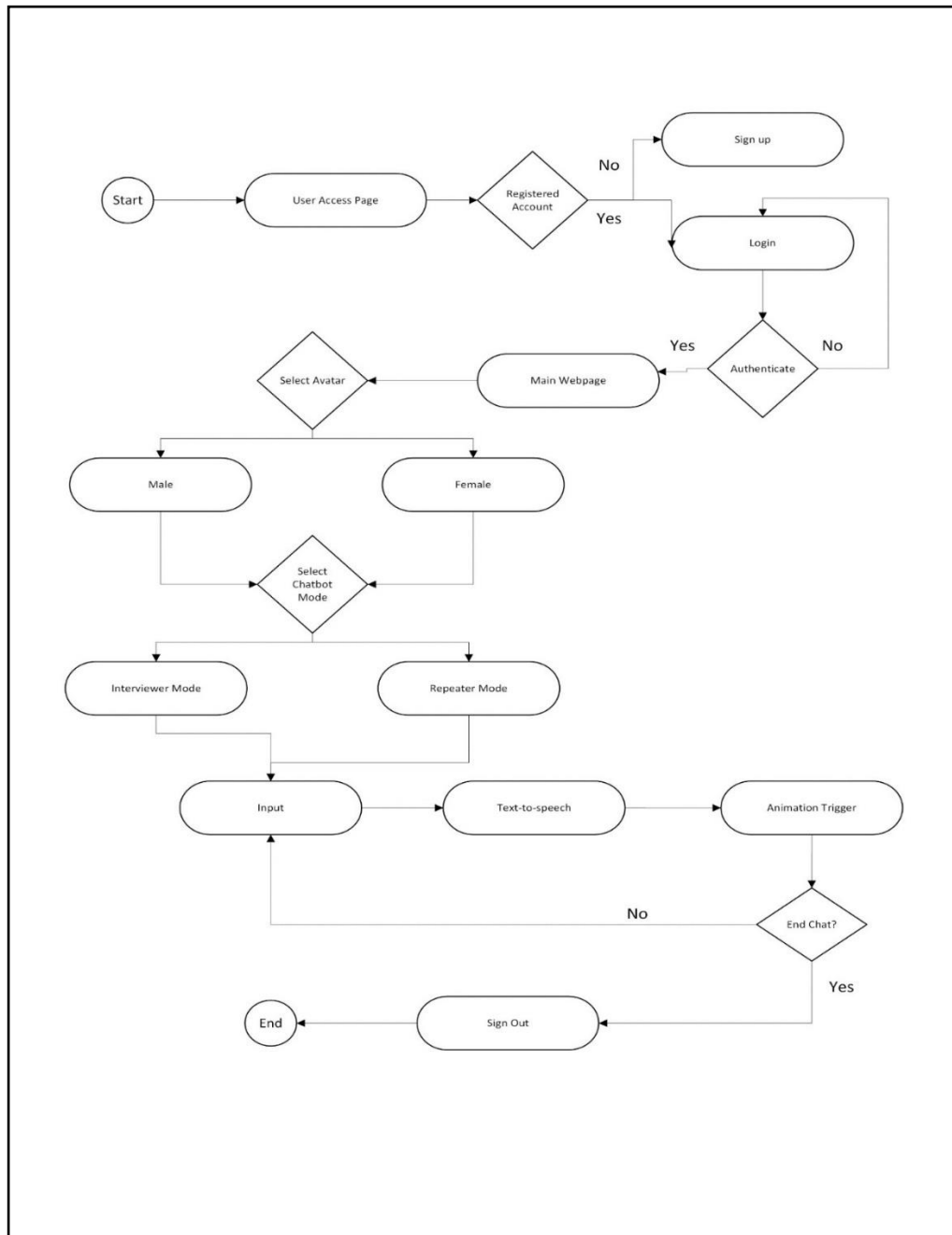
5.3. Development View

5.3.1. Component Diagram

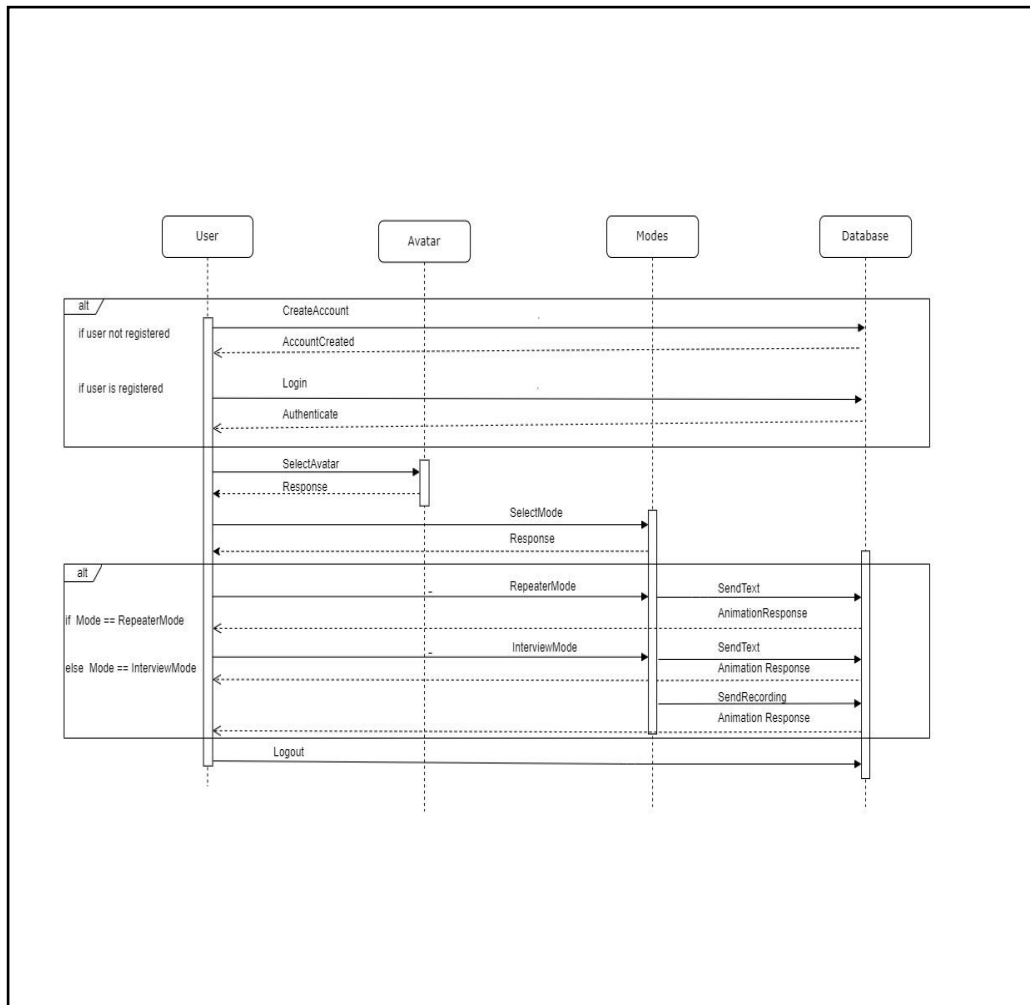


5.4. Process View

5.4.1. Activity Diagram

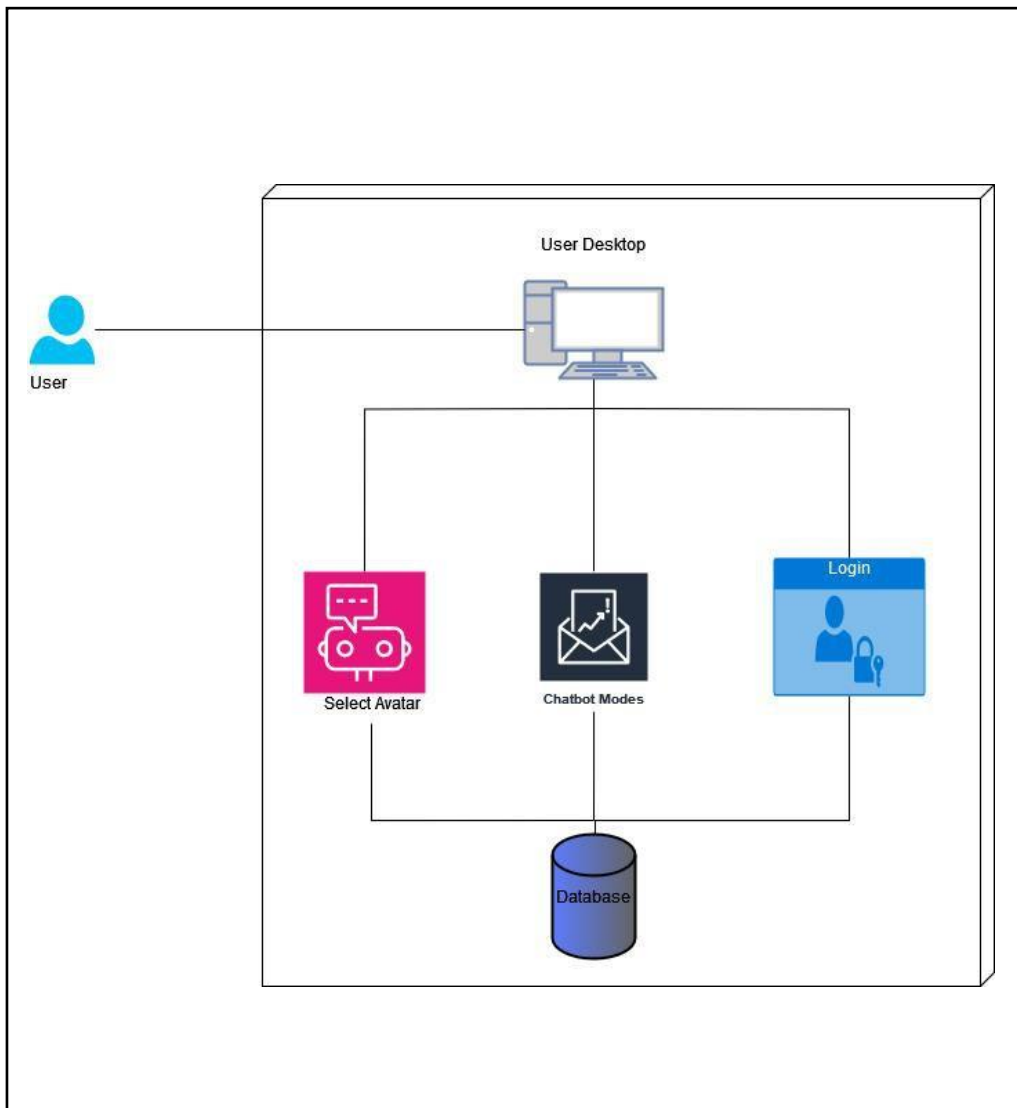


5.4.2. Sequence Diagram



7.5. Physical View

5.4.1. Deployment Diagram



5.6. User Interface Design

5.6.1. Login/Signup Mode



5.6.2. Interview Mode



5.6.2. Repeater Mode



Chapter 6

This chapter provides the outcomes of our project and in-depth discussions, significance, and future findings.

Results and Discussions

By analyzing various existing models and publications we have designed AI-Driven Digital Human Newscaster that redefines the news delivery system and user interaction. The key results and discussions from our project are as follows.

Xin Xiaomeng, ERICA, Lisa, and all other existing models have made significant advancements in the news industry by building the bridge between AI and humans. Sophia, an AI Human-Robot blurs the gap between the humans and robots by incorporation of NLP techniques and human-like interactions. These models offer promising solutions and effective communications in the advancement of the news industry but face certain limitations that are addressed through further research in our project.

By leveraging insights from the last contributions, our system, AI-Driven Digital Human offers users transformative news delivery as well as addresses the challenges in existing models. Our system provides a strong foundation for the development of an AI-Human that addresses biasness in gender and user engagement. The model provides the user with trigger-based interactions to select the gender of the avatar to eliminate bias news delivery to ensure impartiality and inclusivity in the content. By allowing the user to select the gender of the avatar our project ensures fairness of the news dissemination whereas the existing models have predetermined gender which limits the inclusivity. Additionally, our system offers multiple levels of avatar customization, allowing the user to engage and interact. One significant aspect of our system is the incorporation of two modes of chatbot for an immersive user experience. The existing models lack dynamic user engagement, real-time interviews, and multiple modes of chatbot which restricts their ability to interact with the user. These

aspects highlight the innovative approach of our project and how it addresses the challenges in the existing model.

Chapter 7

This chapter highlights the conclusion and future work of our project.

Conclusion & Future Work

AI-Driven Digital Human represents significant advancement in the news industry, by offering engaging user experience through 3D avatars, interactive chatbot modes and human like features. This project redefines the way news is presented and consumed by introducing the selection of customized gender, real-time interviews, multiple dialects, multifaceted operational modes, and different levels of avatars features. This project addresses the key limitations of existing systems and opens up opportunities for future applications. There are several avenues for future enhancements:

To cater the landscape of Pakistan, and to provide user with more customized experience Urdu language can be incorporated to highlight the diverse language preferences. Furthermore, by integrating specialized chatbot models the system can provide versatile experiences and adapt as an Educational Tutor to a Medical Health Advisor.

Chapter 8

This chapter includes the Glossary for our project's report.

Glossary

Table 1: Table for Glossary

AI	Artificial Intelligence
3D	Three-Dimensional
API	Application Programming Interface
RAM	Random-Access Memory
NLP	Natural Language Processing
SST	Speech-to-Text
TTS	Text-to-Speech
UserWidget	Unreal User Widget
MetahumanSDK	Metahuman Software Development Kit
ConvAI	Conversational Artificial
GUI	Graphical User Interface

Chapter 9

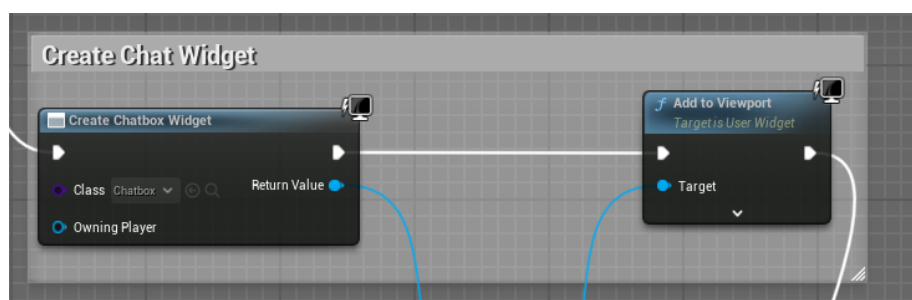
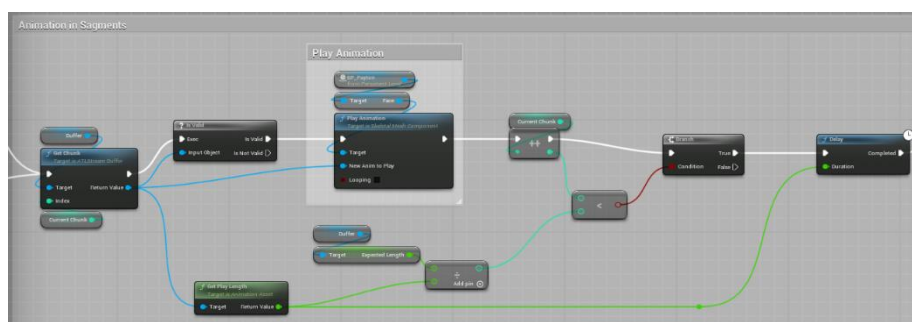
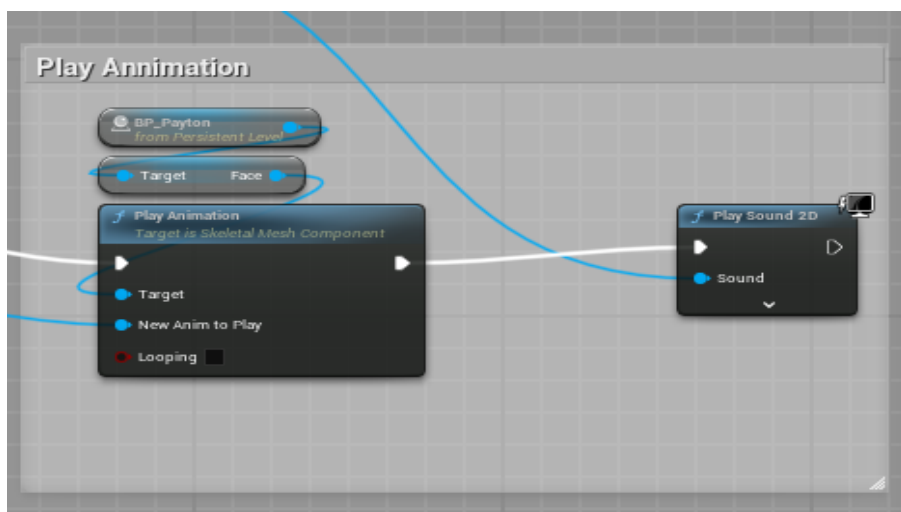
References

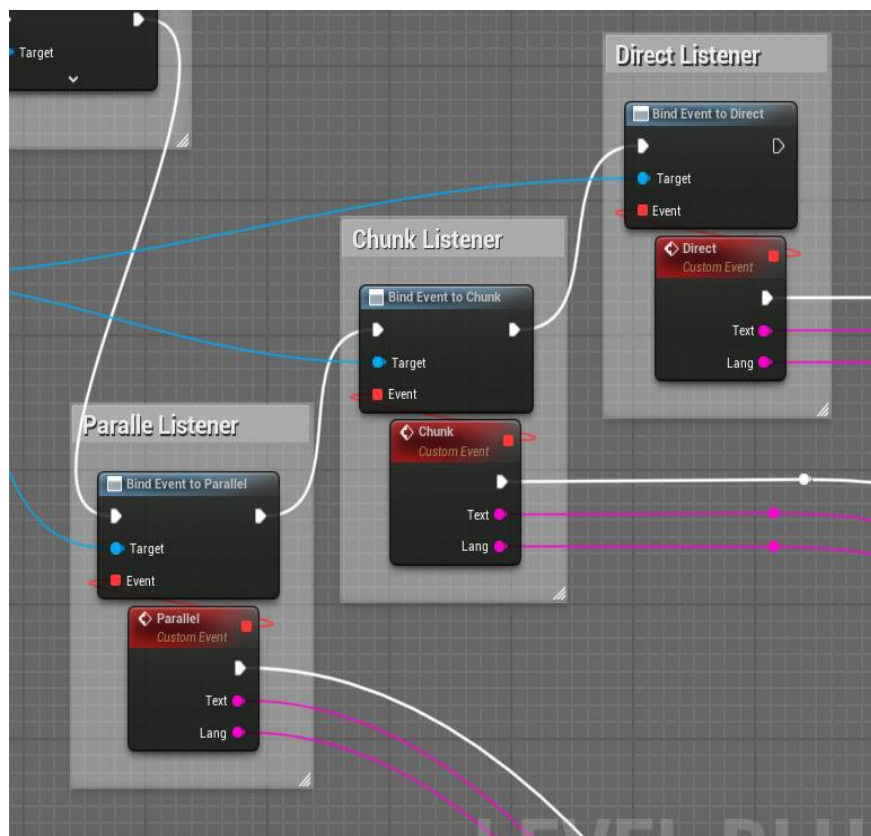
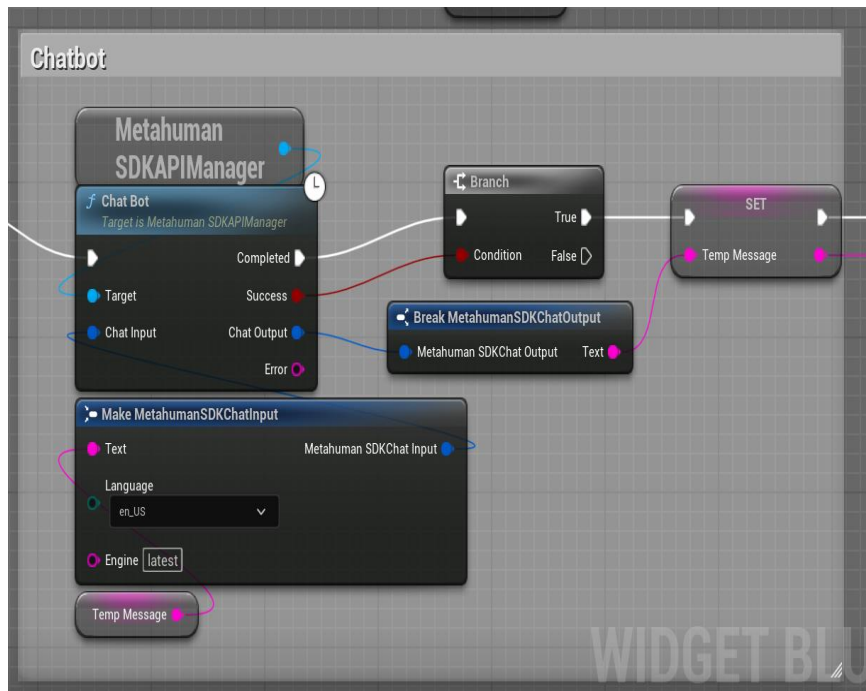
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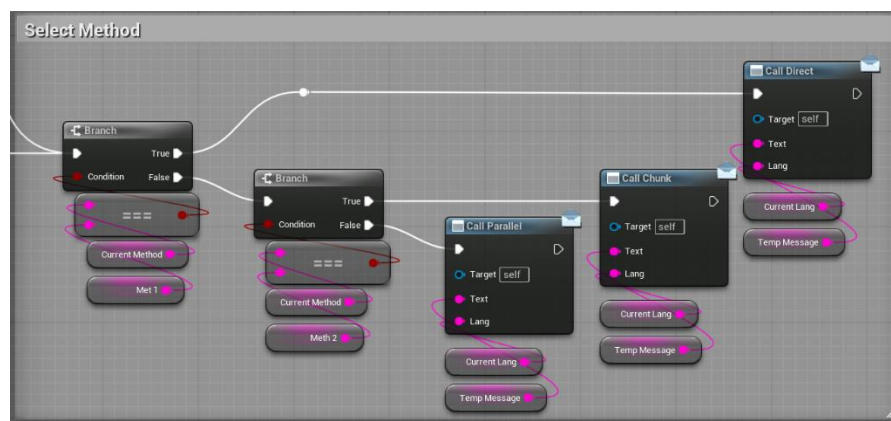
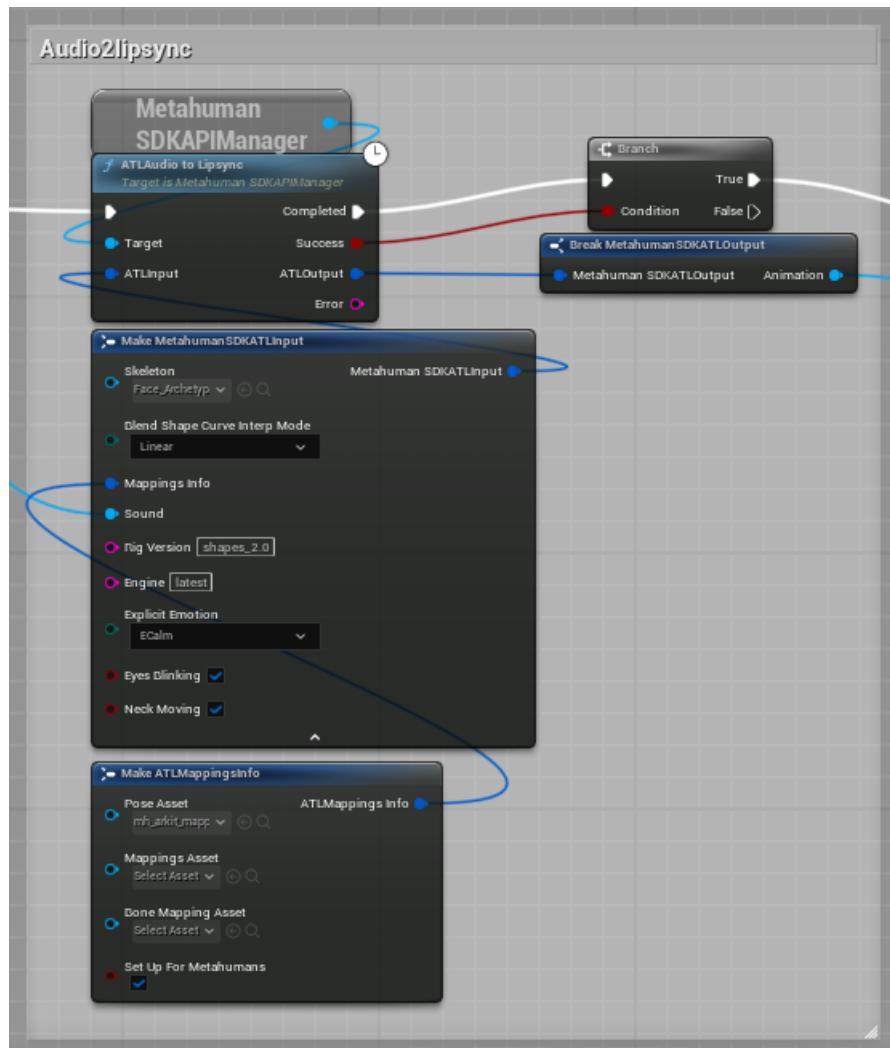
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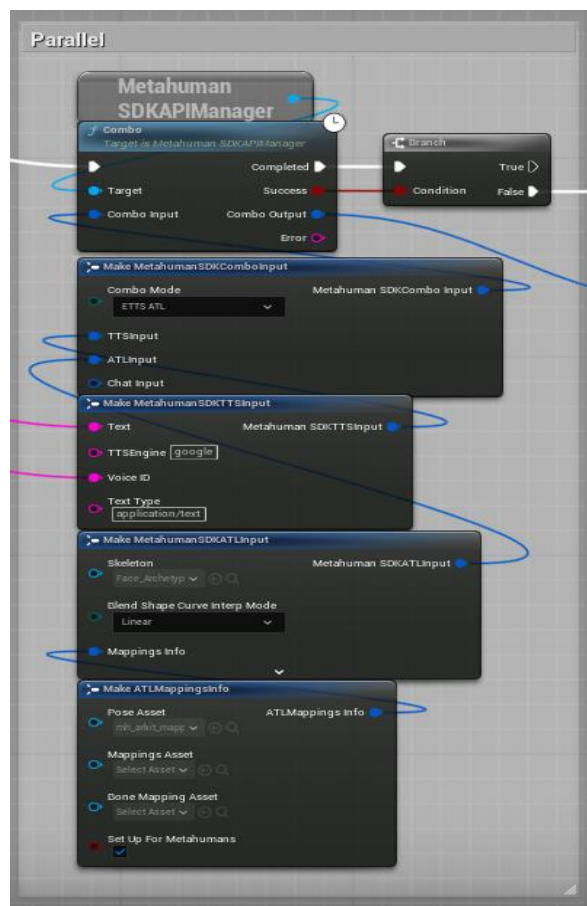
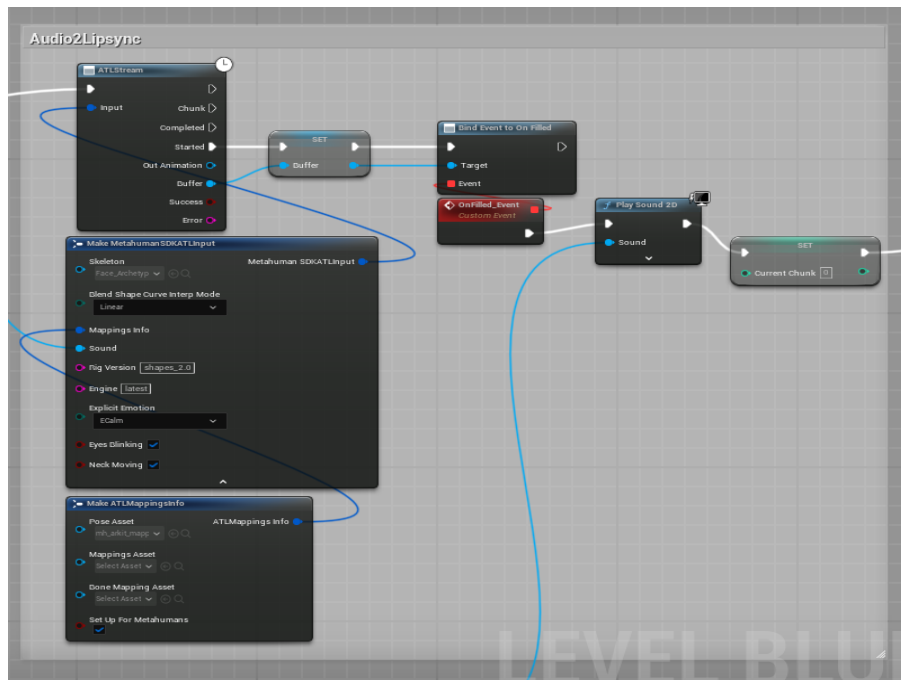
Appendix

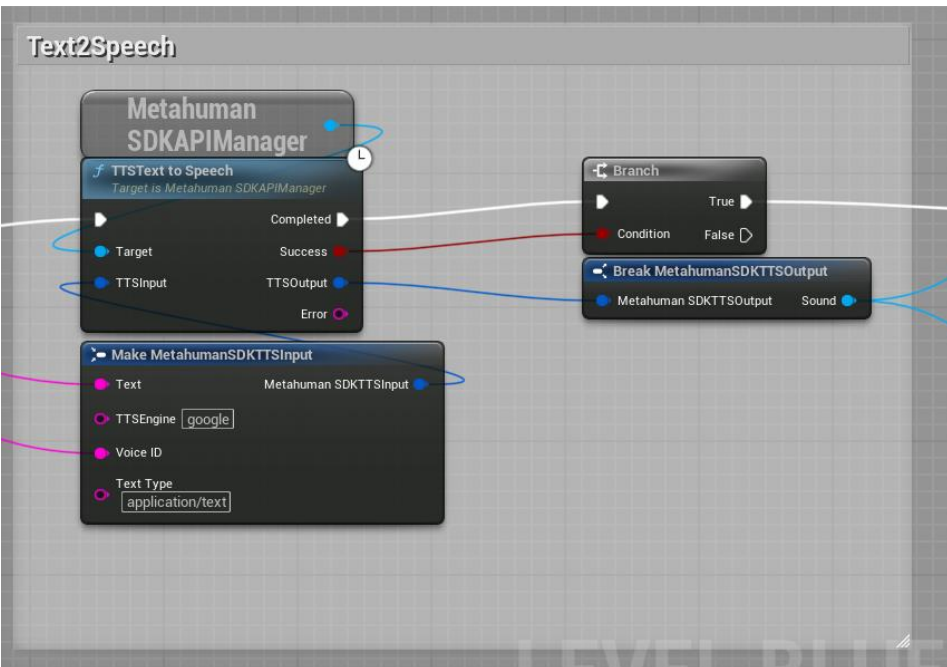
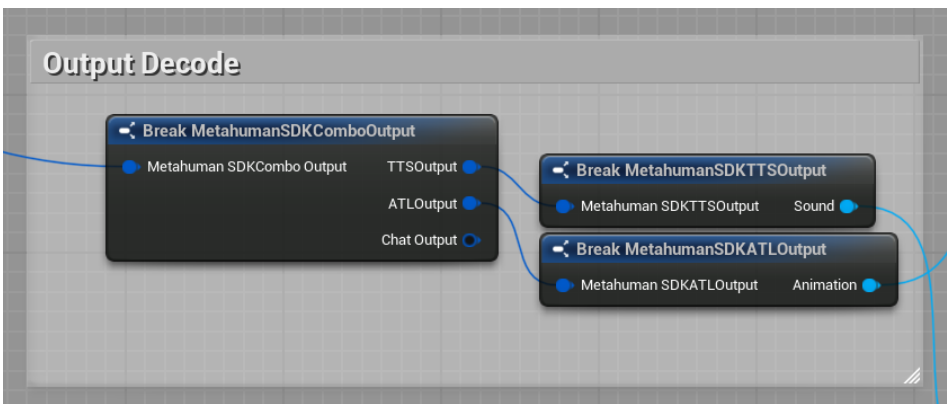
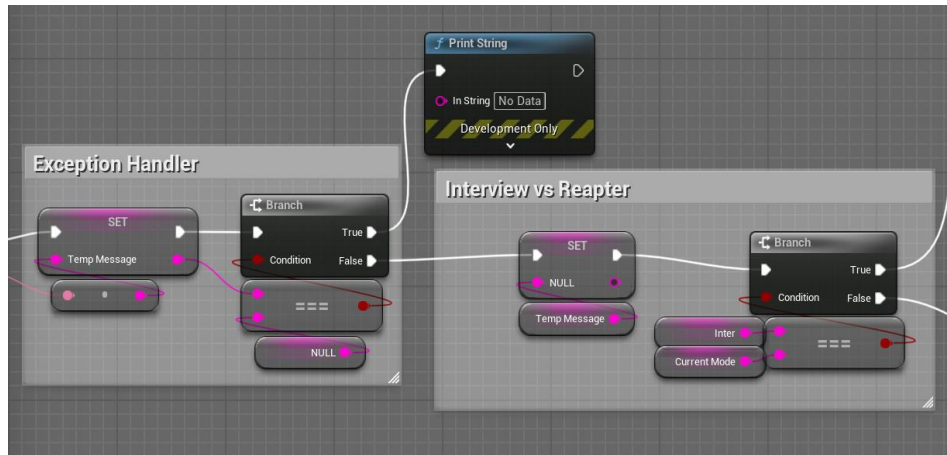
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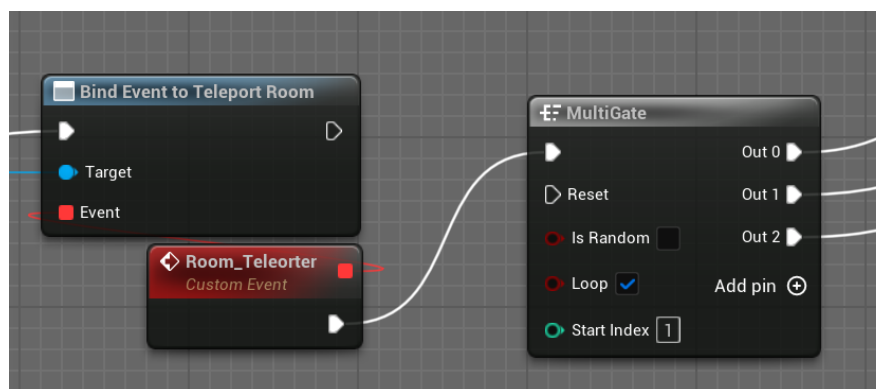
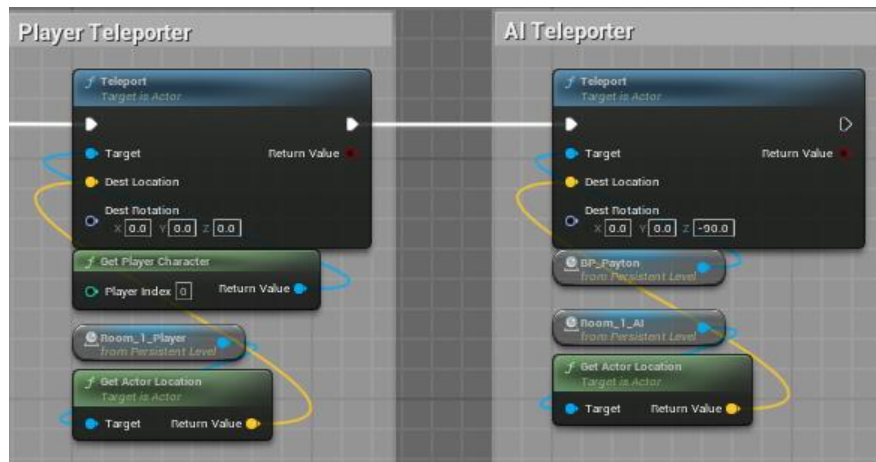






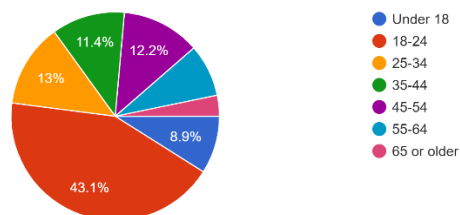




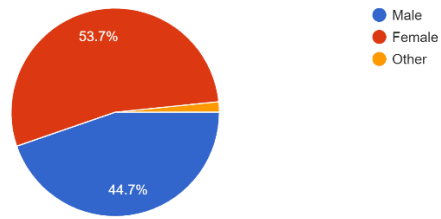


• Survey Form

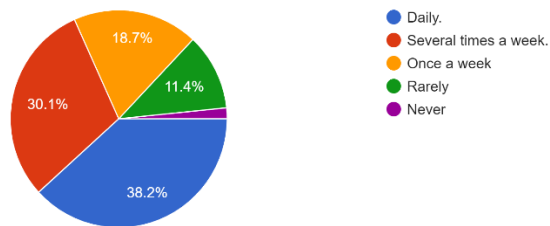
Age:
123 responses



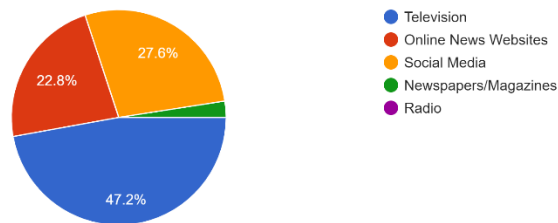
Gender:
123 responses



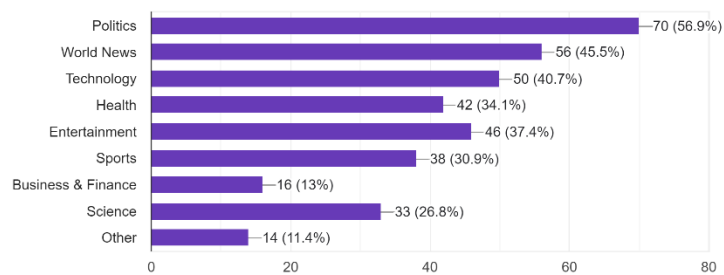
How often do you consume news?
123 responses



Which sources do you primarily use for news?
123 responses

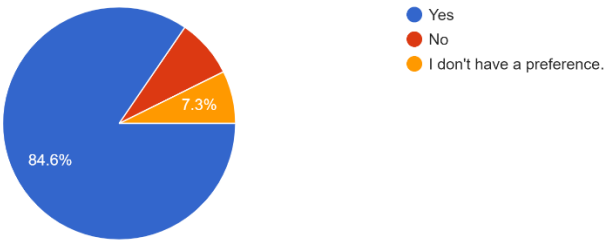


What types of news are you most interested in? (Select all that apply)
123 responses



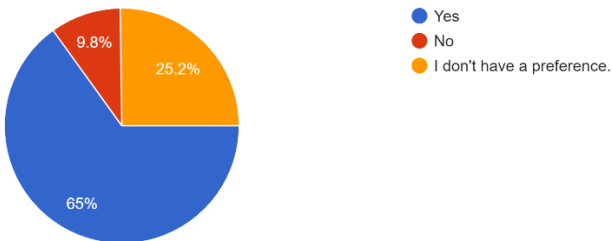
Do you believe that the news industry has traditionally displayed gender bias in the selection of news anchors?

123 responses



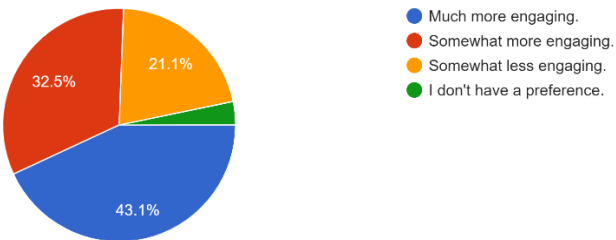
Would you like the option to choose the gender of the newscaster delivering news?

123 responses



Would you find it more engaging to interact with a 3D digital human news anchor compared to watching conventional news broadcasts?

123 responses



AI-DRIVEN DIGITAL HUMAN

ORIGINALITY REPORT

12%

SIMILARITY INDEX

5%

INTERNET SOURCES

1%

PUBLICATIONS

11%

STUDENT PAPERS

PRIMARY SOURCES

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Pakistan

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5%

2

Submitted to Corporación Universitaria
Minuto de Dios, UNIMINUTO

Student Paper

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3

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Student Paper

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